

Ndemiso Baliti

Exercise 13

$$1) i) \frac{\partial u_E}{\partial t} - f v_E = A_v \frac{\partial^2 u_E}{\partial z^2}$$

$$\frac{\partial u_E}{\partial t} = f v_E + A_v \frac{\partial^2 u_E}{\partial z^2}$$

$$u_k^{n+1} = u_k^n + \Delta t \left[ f v_k^n + A_v \left( \frac{u_{k+1}^n - 2u_k^n + u_{k-1}^n}{\Delta z^2} \right) \right]$$

$$ii) \frac{\partial v_E}{\partial t} + f u_E = A_v \frac{\partial^2 v_E}{\partial z^2}$$

$$\frac{\partial v_E}{\partial t} = -f u_E + A_v \frac{\partial^2 v_E}{\partial z^2}$$

$$v_k^{n+1} = v_k^n + \Delta t \left[ -f u_k^n + A_v \left( \frac{v_{k+1}^n - 2v_k^n + v_{k-1}^n}{\Delta z^2} \right) \right]$$

$$2.) \frac{\partial u_E}{\partial t} = f v_E + A_v \frac{\partial^2 u_E}{\partial z^2}$$

$$\frac{\partial^2 u_E}{\partial z^2} = \frac{u_{k+1}^n - 2u_k^n + u_{k-1}^n}{\Delta z^2}$$

$$\frac{\partial u_E}{\partial t} = f v_k^n + A_v \left( \frac{u_{k+1}^n - 2u_k^n + u_{k-1}^n}{\Delta z^2} \right)$$

$$\frac{u_k^{n+1} - u_k^n}{\Delta t} = f v_k^n + A_v \left( \frac{u_{k+1}^n - 2u_k^n + u_{k-1}^n}{\Delta z^2} \right)$$

$$u_k^{n+1} - u_k^n = \Delta t f v_k^n + A_v \Delta t \left( \frac{u_{k+1}^n - 2u_k^n + u_{k-1}^n}{\Delta z^2} \right)$$

$$u_k^{n+1} = u_k^n + \Delta t f v_k^n + \frac{A_v \Delta t}{\Delta z^2} (u_{k+1}^n - 2u_k^n + u_{k-1}^n)$$

$$C_v = \frac{A_v \Delta t}{\Delta z^2}$$

3.)  $\Delta z = 1\text{m}$  and  $\Delta t = 1000\text{s}$ ,  
because it gives stability

$$\frac{10^{-4} \times 1000}{1.2} = 0.1$$

it is below 0.5 so the solution should be stable.

$$\frac{30 \pm 5}{2 \times 6} \Delta t + 3V = \frac{3 \pm 6}{2}$$

$$\left[ \frac{30 \pm 5}{2 \times 6} \Delta t + 3V \right] \Delta t + 3V = \frac{3 \pm 6}{2}$$

$$\frac{30 \pm 5}{2 \times 6} \Delta t = 3V + \frac{3 \pm 6}{2}$$